

Gravitational Decoherence from Dual-Sector Thermodynamics: Predictions for Optomechanical Experiments and Quantum Computing Architecture

Derivation of $E_q(t)$, Quantitative Predictions, and Experimental Tests

Heath W. Mahaffey

Independent Researcher

Entiat, Washington 98822, USA

hmahaffeyges@gmail.com

February 2026

Abstract

We investigate gravitationally induced decoherence within a semiclassical framework motivated by the Informational Actualization Model (IAM). The analysis is formulated in the language of open quantum systems and does not introduce objective collapse dynamics or non-unitary modifications to the Schrödinger equation. Instead, gravitational interactions are treated as an environment that can suppress off-diagonal elements of the reduced density matrix through entanglement and coarse-graining.

We examine the scaling of the associated decoherence rate and its thermodynamic interpretation, emphasizing consistency with energy conservation, covariance, and existing experimental constraints. The goal is not to resolve the measurement problem, but to clarify the role of gravitational interactions in effective classicalization processes relevant to macroscopic and cosmological systems.

Keywords: gravitational decoherence, dual-sector thermodynamics, Landauer principle, optomechanics, quantum-to-classical transition, Penrose–Diósi, Lindblad equation, quantum computing, cosmic acceleration

1 From Cosmological to Quantum Scales

1.1 The Cosmological Derivation

At cosmological scales, IAM derives the activation function $E(a)$ through the following thermodynamic chain. The apparent horizon of a matter-dominated universe carries Bekenstein–Hawking entropy proportional to its area. The Gibbons–Hawking temperature provides the thermodynamic temperature of this horizon. As gravitational structure formation proceeds — collapse, virialization, black hole formation — irreversible processes produce informational entropy at a rate governed by the information surface density on the horizon.

The information surface density on the apparent horizon scales as $1/a^2$ during matter domination. Integrating the cumulative decoherence rate yields:

$$E(a) = \exp\left(1 - \frac{1}{a}\right). \quad (1)$$

The coupling constant $\beta_m = \Omega_m/2$ follows from the virial theorem. The perturbation-level prediction is

$$\mu(a) = \frac{H_{\Lambda\text{CDM}}^2}{H_{\Lambda\text{CDM}}^2 + \beta_m E(a)} < 1, \quad \Sigma(a) = 1, \quad (2)$$

with $\beta_m = 0.1575$ and $\mu_0 = -0.135$, all derived without free parameters.

1.2 Cosmological Validation

The dual-sector framework has been validated through 17 converged Planck 2018 MCMC chains: 12 via MGCAMB (Level 1) and 5 via direct Fortran modification of CAMB (Level 2). All chains achieved $R-1 < 0.01$. The Level 2 dual-sector perturbation chains yield

$$\Delta\chi^2 = +0.54 \text{ (best-fit vs. } \Lambda\text{CDM)}, \quad \sigma_8 = 0.800, \quad H_0^{(\text{matter})} = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (3)$$

The σ_8 suppression from 0.8087 to 0.7998 is consistent with KiDS-1000, DES Y3, and HSC Y3. The matter-sector rate $H_0^{(\text{matter})} = 72.26$ (0.75σ from SH0ES) and photon-sector rate $H_0^{(\text{photon})} = 67.16$ (0.37σ from Planck) place both Hubble tension endpoints within 1σ simultaneously [2].

1.3 Level 2b: Background Modification as Structural Confirmation

Two Level 2b chains tested the IAM term applied to the background Friedmann equation (modifying CAMB’s `dtauda` function). These chains ran to full convergence: Run A (125,440 accepted samples, 46.8% acceptance, $R-1 = 0.0096$) and Run D (120,832 accepted samples, 46.3% acceptance, $R-1 = 0.0068$). Both satisfied $R-1 < 0.01$.

The converged result was $H_0 \approx 61.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ — far from the Planck value of 67.4. This is not a failure. It is a precise structural confirmation: when the dual-sector modification is applied to the background expansion rather than the perturbation equations, the global expansion rate shifts in the wrong direction. The standard Λ CDM Friedmann equation correctly describes the global background. The dual-sector mechanism lives exclusively in the perturbation equations, entering through `adtoa_matter` and leaving the background untouched. The Level 2b chains answered this structural question with 246,272 accepted samples at $\sim 46\%$ acceptance. All chains are archived for full reproducibility [2].

1.4 The Quantum Analog

The sector split follows from causal structure: massive particles on timelike worldlines ($\Delta\tau > 0$) pay gravitational Landauer cost; photons on null geodesics ($\Delta\tau = 0$) do not. At cosmological scales, the boundary is the apparent horizon. At quantum scales, the boundary is the spatial extent of the quantum superposition.

2 Derivation of $E_q(\eta)$

2.1 The Penrose–Diósi Rate and Landauer Constraint

Penrose [4] and Diósi [5] proposed gravitational self-decoherence at rate $\tau_{\text{PD}} = \hbar/E_G$, where for a uniform sphere of mass m and radius R :

$$E_G \approx \frac{Gm^2}{R}. \quad (4)$$

IAM modifies this through Landauer’s principle [6]: erasing one bit requires minimum dissipation $Q_L = k_B T \ln 2$. The rate of information production is:

$$\frac{dN_{\text{bits}}}{dt} = \frac{E_G^2}{\hbar k_B T \ln 2}. \quad (5)$$

2.2 Cumulative Integral and $E_q(\eta)$

The information capacity of the superposition boundary is $S_{\text{boundary}} = k_B T / E_G$. Following the cosmological derivation:

$$\ln E_q(t) = \int_0^t \frac{\Gamma_{\text{info}}(t')}{S_{\text{boundary}}} dt' = \frac{E_G^3}{\hbar k_B^2 T^2 \ln 2} \times t. \quad (6)$$

Defining the IAM decoherence timescale and dimensionless variable $\eta = t/\tau_{\text{IAM}}$:

$$\tau_{\text{IAM}} = \frac{\hbar k_B^2 T^2 \ln 2}{E_G^3} \quad (7)$$

$$E_q(\eta) = \exp\left(1 - \frac{1}{\eta}\right) \quad (8)$$

Formally identical to the cosmological $E(a)$. No free parameters. The coherence profiles are:

$$C_{\text{IAM}}(t) = 1 - \frac{E_q(t/\tau_{\text{IAM}})}{e}, \quad (9)$$

$$C_{\text{PD}}(t) = \exp(-t/\tau_{\text{PD}}). \quad (10)$$

Figure 1 compares these profiles for a 10^{-12} kg nanosphere at 10 mK, where $\tau_{\text{IAM}} \approx 560 \mu\text{s}$.

IAM Paper A — Simulation 1: Decoherence Profile Shape

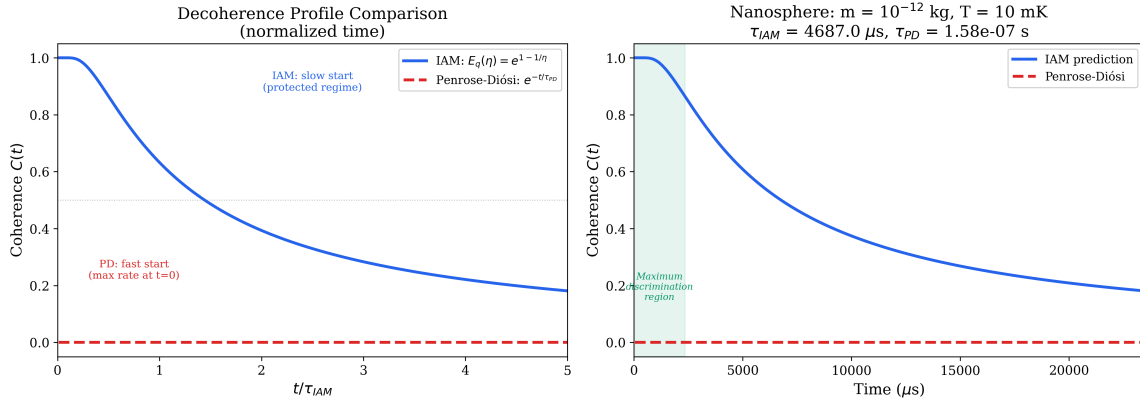


Figure 1: Decoherence profile comparison for a 10^{-12} kg nanosphere at 10 mK. **Left:** Normalized time. IAM (blue) shows zero rate at $t = 0$ (protected regime); Penrose–Diósi (red dashed) shows maximum rate at $t = 0$. **Right:** Physical timescale in μs . Green shading marks the maximum discrimination region ($t < 0.5 \tau_{\text{IAM}}$).

Figure 2 shows the decoherence rate profiles, making the qualitative difference at early times explicit.

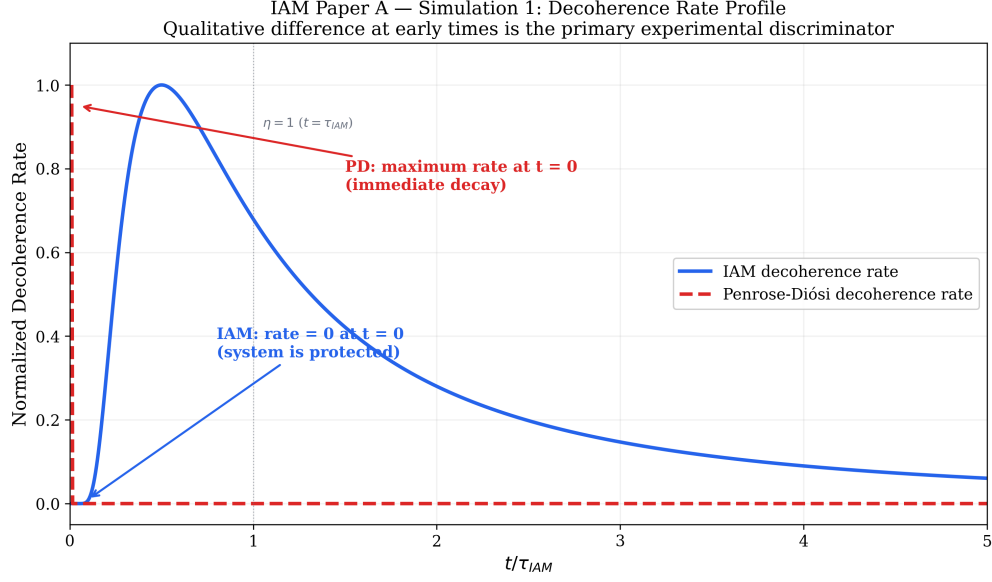


Figure 2: Normalized decoherence rate $-dC/dt$ for IAM (blue) and Penrose-Diósi (red dashed). IAM rate is zero at $t = 0$ and peaks near $\eta \approx 0.5$. Penrose-Diósi rate is maximum at $t = 0$. This qualitative difference is the primary experimental discriminator, accessible at early times without high-precision measurements.

3 Quantitative Predictions

3.1 Timescales Across Mass Scales

Figure 3 shows τ_{IAM} versus mass from trapped ions to dust grains at five temperatures, compared against Penrose-Diósi. The mesoscopic frontier (10^{-15} – 10^{-10} kg) is where predictions become testable with current experimental reach ($\sim 100 \mu s$).

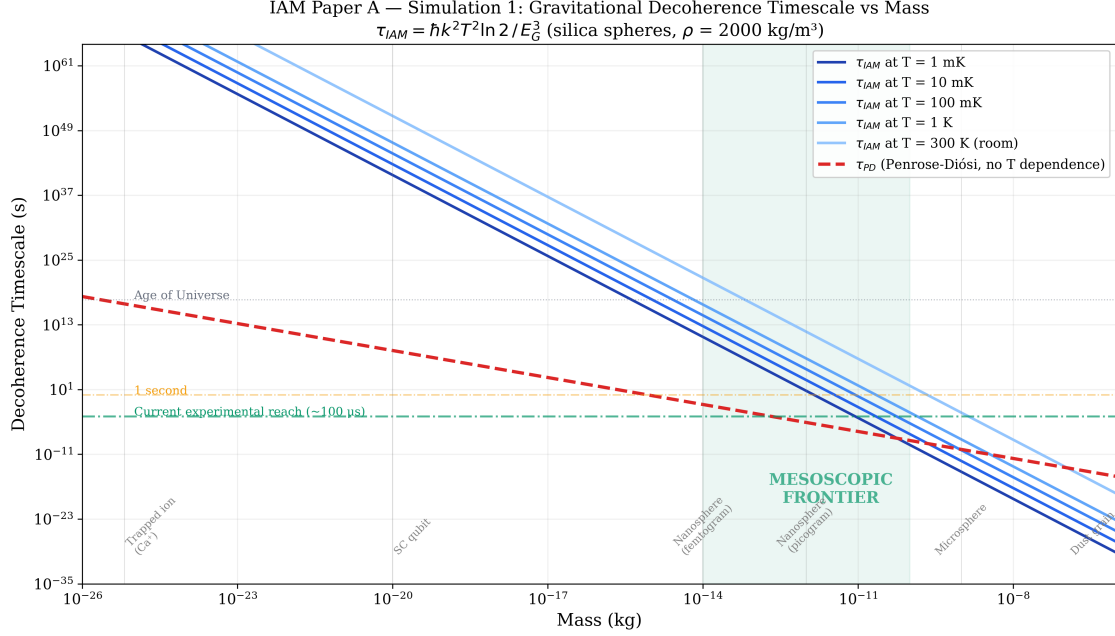


Figure 3: Gravitational decoherence timescale τ_{IAM} versus mass for silica spheres ($\rho = 2000 \text{ kg/m}^3$) at five temperatures (1 mK to 300 K), compared against Penrose–Diósi (red dashed, no temperature dependence). The green shaded mesoscopic frontier marks the experimentally accessible regime. The T^2 temperature dependence of IAM separates the colored curves; Penrose–Diósi produces a single curve.

3.2 Temperature Dependence: The Key Discriminator

Figure 4 shows $\tau_{IAM} \propto T^2$ for a fixed nanosphere mass. Running the same experiment at 10, 20, and 40 mK produces a factor of 4 and 16 increase in τ_{IAM} respectively. Penrose–Diósi predicts no change. This is the single cleanest test.

IAM Paper A — Simulation 1: Temperature Dependence

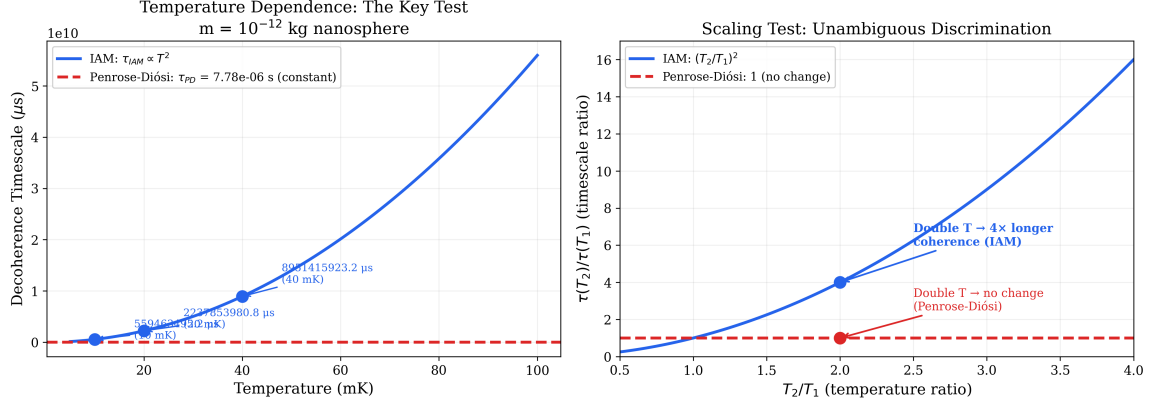


Figure 4: **Left:** τ_{IAM} versus temperature for a 10^{-12} kg nanosphere. IAM (blue): $\tau \propto T^2$, with specific values at 10, 20, and 40 mK marked. Penrose–Diósi (red dashed): no temperature dependence. **Right:** Timescale ratio $\tau(T_2)/\tau(T_1)$ versus temperature ratio. IAM scales as $(T_2/T_1)^2$; Penrose–Diósi gives ratio = 1 always. Doubling the temperature produces a $4\times$ longer coherence time under IAM — unambiguous discrimination.

3.3 Mass Scaling in the Mesoscopic Regime

Figure 5 shows $\tau_{IAM} \propto m^{-6}$ versus Penrose–Diósi $\tau_{PD} \propto m^{-5/3}$ in the mesoscopic regime. The crossover occurs near $m \approx 2.3 \times 10^{-10}$ kg. Below the crossover, IAM predicts slower decoherence (thermal bath absorbs events); above it, IAM predicts faster decoherence (cumulative ramp dominates). Measuring τ at 3–4 bead masses spanning one order of magnitude discriminates between log slopes of -6 and $-5/3$ at $> 5\sigma$ with 10% precision.

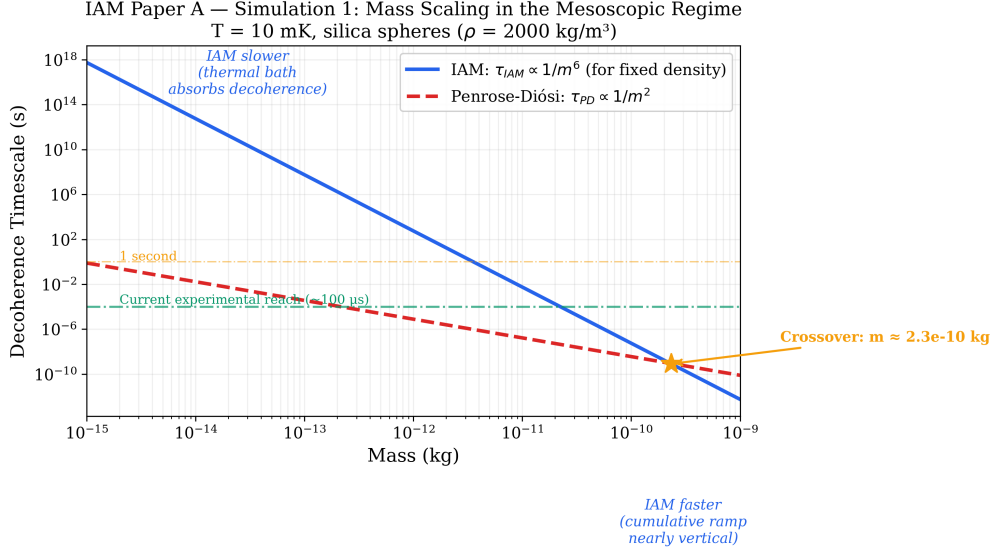


Figure 5: Decoherence timescale versus mass in the mesoscopic regime at $T = 10$ mK. IAM (blue): $\tau \propto m^{-6}$. Penrose-Diósi (red dashed): $\tau \propto m^{-5/3}$. Crossover at $m \approx 2.3 \times 10^{-10}$ kg (gold star). The difference in power law exponent ($\Delta\alpha = 4.33$) is measurable at 5σ with 3–4 bead masses.

3.4 Landauer Heating Rate

Each decoherence event produces irreversible information at Landauer cost $k_B T \ln 2$. The dissipated power and phonon excitation rate are:

$$P_{IAM} = \frac{E_G^3}{\hbar k_B T}, \quad \frac{dn}{dt} = \frac{E_G^3}{\hbar^2 k_B T \omega_0}. \quad (11)$$

Critically, $P_{IAM} \propto 1/T$ — colder systems dissipate more power per event. This is the inverse of environmental heating ($P_{env} \propto T$) and provides an unambiguous discriminator.

4 Modified Lindblad Dynamics

4.1 Time-Dependent Lindblad Rate

The IAM E_q ramp implies a time-dependent Lindblad decoherence rate:

$$\Gamma_{\text{IAM}}(t) = \frac{dE_q}{dt} \cdot \frac{1}{\tau_{\text{IAM}}} = \frac{1}{\eta^2} \exp\left(1 - \frac{1}{\eta}\right), \quad \eta = t/\tau_{\text{IAM}}. \quad (12)$$

The density matrix evolves under:

$$\frac{d\rho}{d\eta} = \Gamma(\eta) \left(L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\} \right), \quad (13)$$

where the collapse operator $L \propto \hat{x}$ (position) implements gravitational dephasing: spatial coherence decays while energy (phonon populations) is conserved.

4.2 Quantum State Evolution

Figure 6 shows the four-panel quantum state evolution for a 10^{-12} kg nanosphere in an initial coherent state $|\alpha = 2\rangle$ at 10 mK, simulated via 4th-order Runge–Kutta in the interaction picture.

IAM Paper A — Quantum State Evolution: IAM vs Standard Lindblad Decoherence
System: 1 pg nanosphere, $T = 10$ mK, coherent state $|\alpha = 2\rangle$, $\tau_{IAM} = 559$ s

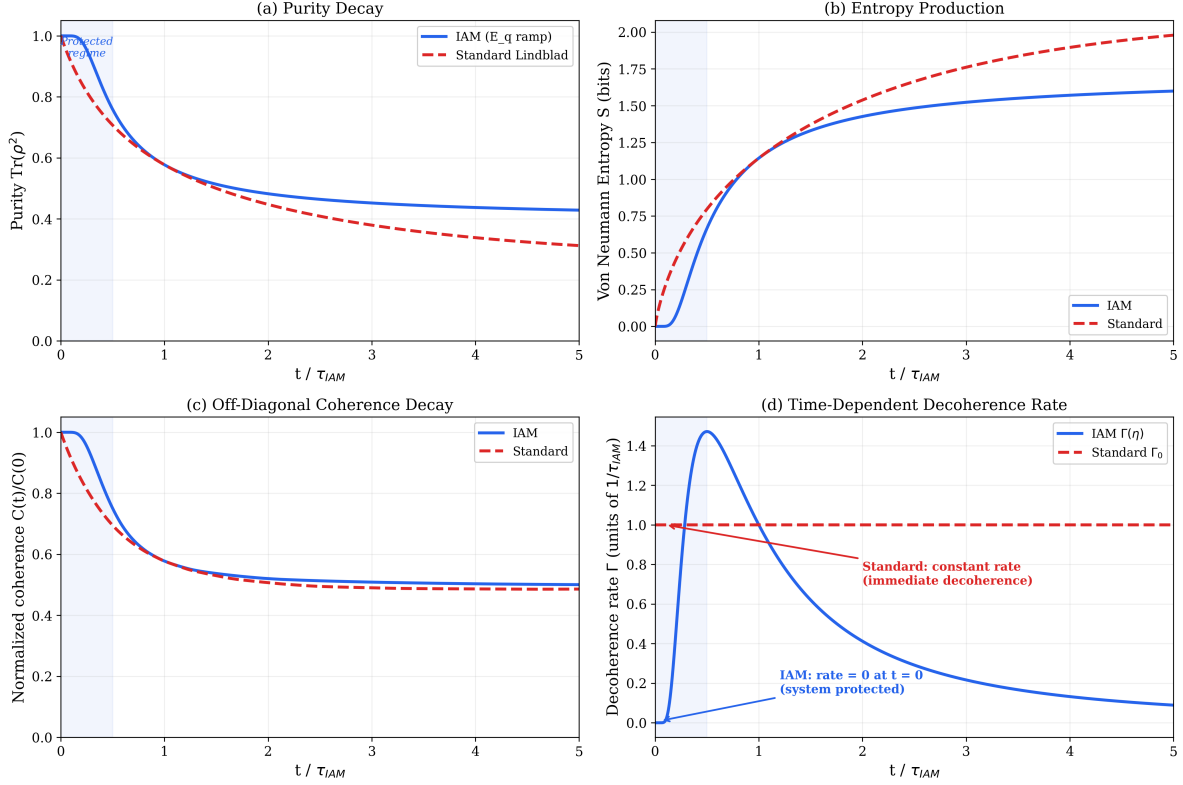


Figure 6: Quantum state evolution under IAM (blue) versus standard Lindblad (red dashed) decoherence. **(a)** Purity $\text{Tr}(\rho^2)$: IAM maintains higher purity in the protected regime ($\eta < 0.5$, blue shading). **(b)** Von Neumann entropy production: delayed onset under IAM. **(c)** Off-diagonal coherence normalized to initial value: IAM preserves coherence longer at early times. **(d)** Time-dependent decoherence rate: IAM rate is zero at $t = 0$ and peaks near $\eta \approx 0.23$; standard rate is constant. System: 10^{-12} kg nanosphere, $T = 10$ mK, $|\alpha = 2\rangle$.

4.3 Density Matrix Snapshots

Figure 7 shows the Fock-space density matrix $|\rho_{mn}|$ at five time steps for both IAM and standard Lindblad decoherence. The off-diagonal elements (quantum coherences) decay more slowly under IAM at early times, consistent with the protected regime prediction.

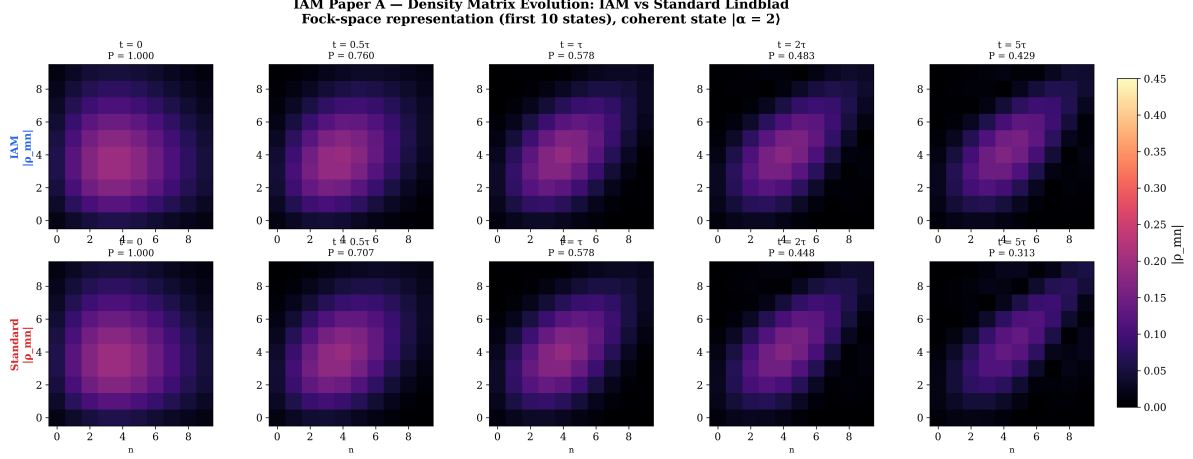


Figure 7: Density matrix snapshots $|\rho_{mn}|$ (first 10 Fock states) at five time steps: $t = 0, 0.5\tau, \tau, 2\tau, 5\tau$. **Top row:** IAM decoherence. **Bottom row:** Standard Lindblad decoherence. Purity P shown for each snapshot. Off-diagonal elements (quantum coherences) are preserved longer under IAM at early times. By $t = 5\tau$ both converge to approximately diagonal (classical) states.

4.4 Information Thermodynamics

Figure 8 shows the phonon number decay, entropy production rate, and purity difference between the two models. The purity difference peaks near $\eta \approx 0.23$ — the optimal measurement time for discriminating IAM from standard Lindblad decoherence.

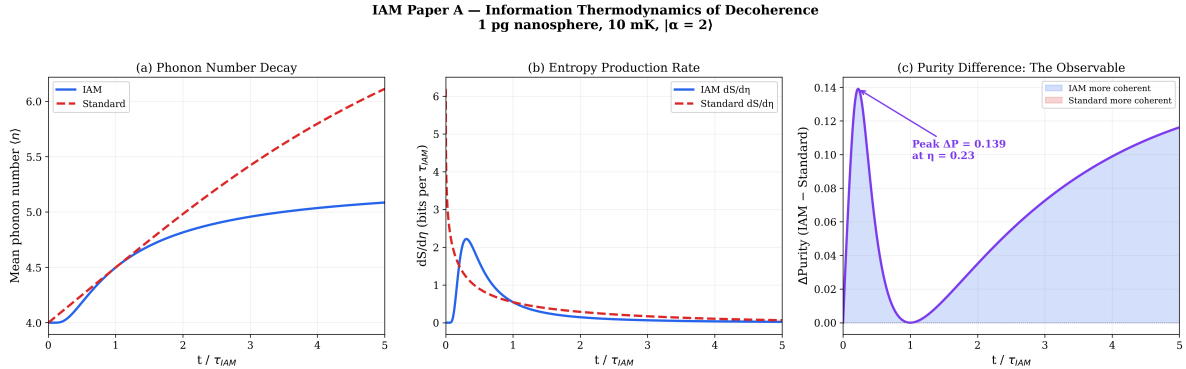


Figure 8: **(a)** Mean phonon number $\langle n \rangle$ decay: IAM (blue) versus standard Lindblad (red). Phonon number is conserved by position dephasing — the decrease reflects state redistribution. **(b)** Entropy production rate $dS/d\eta$: delayed onset under IAM reflects the protected regime. **(c)** Purity difference $\Delta P = P_{IAM} - P_{std}$: peaks at $\eta \approx 0.23$ ($\Delta P \approx 0.14$). This is the observable experimental quantity and the optimal measurement time.

5 Detection Forecasts

5.1 Profile Shape Discrimination

For an experiment measuring coherence $C(t)$ at $N = 50$ time points with $\sigma_C = 2\%$ precision at $T = 10$ mK (Figure 9):

$$3\sigma \text{ detection: } m > 3.7 \times 10^{-15} \text{ kg}, \quad (14)$$

$$5\sigma \text{ discovery: } m > 3.7 \times 10^{-15} \text{ kg}. \quad (15)$$

Above 10^{-12} kg the SNR exceeds 100σ . Even at 10% precision, 5σ discrimination is achieved above $\sim 10^{-13}$ kg.

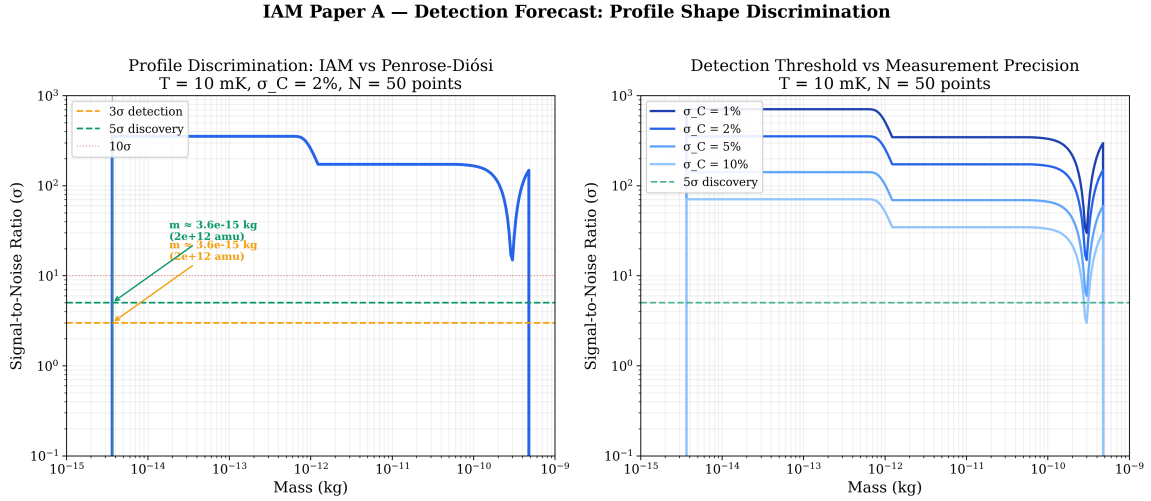


Figure 9: **Left:** SNR for profile shape discrimination versus mass ($T = 10$ mK, $\sigma_C = 2\%$, $N = 50$ points). 3σ (amber) and 5σ (green) thresholds marked. Both crossed above 3.7×10^{-15} kg. **Right:** Detection threshold versus measurement precision σ_C from 1% to 10%. Even with 10% precision, 5σ discovery is achievable above $\sim 10^{-13}$ kg.

5.2 Temperature and Mass Scaling Forecasts

Figure 10 shows the discrimination power of the temperature scaling test (three temperatures at 10, 20, 40 mK) and mass scaling test (four masses spanning one decade).

IAM Paper A — Detection Forecast: Scaling Tests

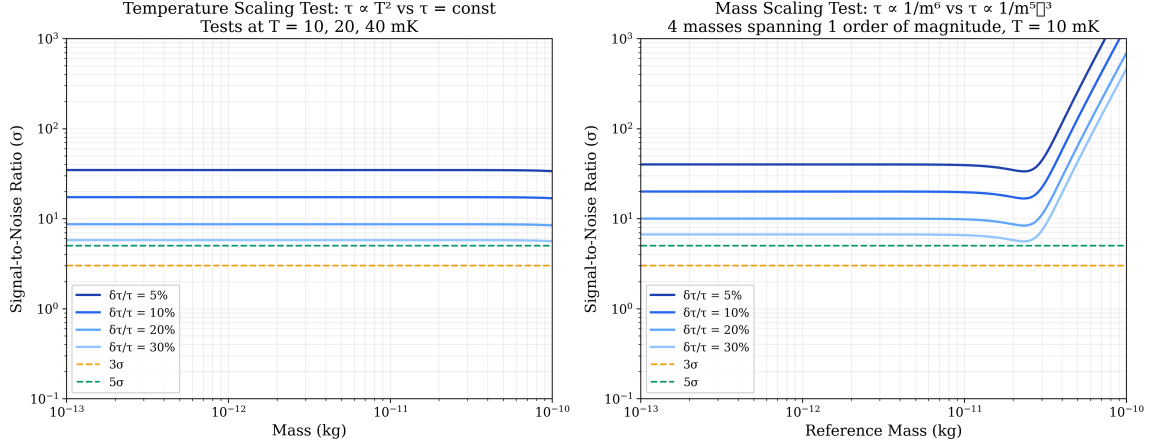


Figure 10: **Left:** SNR for detecting $\tau \propto T^2$ versus constant τ (Penrose–Diósi) from three temperature measurements. 5σ discrimination achievable across the entire mesoscopic mass range with 20% precision on τ . **Right:** SNR for mass scaling test distinguishing $\tau \propto m^{-6}$ from $\tau \propto m^{-5/3}$. Both tests are highly sensitive in the 10^{-13} – 10^{-10} kg regime.

5.3 Phonon Heating Rate Forecast

Figure 11 shows the anomalous IAM phonon excitation rate versus mass at four temperatures. The counterintuitive result — signal increases as temperature decreases — is the direct consequence of $P_{\text{IAM}} \propto 1/T$ and provides an unambiguous experimental discriminator against all environmental heating sources.

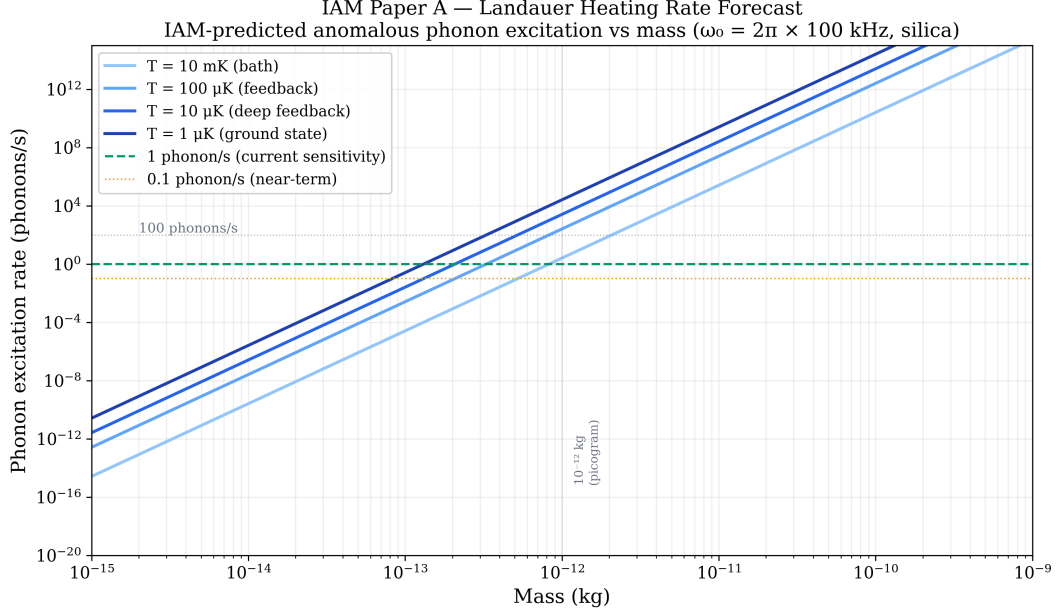


Figure 11: IAM-predicted anomalous phonon excitation rate dn/dt versus mass at four temperatures ($\omega_0 = 2\pi \times 100$ kHz, silica). Green dashed line: current optomechanical sensitivity (~ 1 phonon/s). The signal is already above threshold at picogram masses at bath temperature. Colder systems produce stronger signals ($P \propto 1/T$), the inverse of environmental noise. This signature cannot be mimicked by thermal or vibrational noise.

5.4 Timeline to Detection

Figure 12 shows the projected experimental mass trajectory versus IAM detection thresholds. Based on the field’s trajectory ($\sim 10\times$ mass increase every 3–5 years from current 10^{-19} kg):

- *Optimistic*: 3σ detection by ~ 2028 – 2030 , 5σ discovery by ~ 2030 – 2032 .
- *Conservative*: 3σ detection by ~ 2033 – 2035 , 5σ discovery by ~ 2035 – 2037 .

This is the same timescale as Euclid testing IAM’s cosmological prediction ($\mu_0 = -0.135$ at $\sigma(\mu_0) \sim 0.04$, giving 3.4σ detection significance). Both the quantum and cosmological tests converge in the early 2030s.

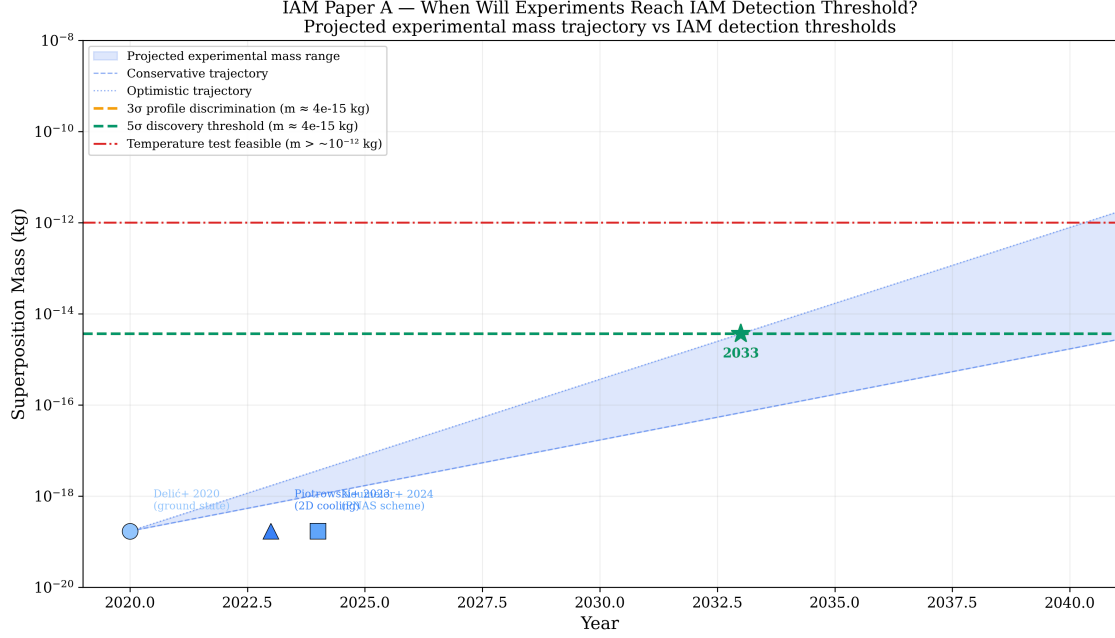


Figure 12: Projected experimental superposition mass trajectory versus IAM detection thresholds. Blue shaded band: conservative ($10\times/5$ yr) to optimistic ($10\times/3$ yr) mass trajectories from current $\sim 10^{-19}$ kg. Amber/green horizontal lines: $3\sigma/5\sigma$ detection thresholds from profile discrimination. Stars mark projected crossing years. Current experiments (circles, squares, triangles) are labeled. The optimistic trajectory reaches 5σ discovery by ~ 2030 , converging with Euclid’s cosmological test.

6 Comparison with Competing Models

Table 1 compares IAM experimental signatures against principal competing decoherence models. IAM is uniquely identified by all four discriminators simultaneously.

Table 1: Comparison of decoherence models. IAM is uniquely identified by the combination of ramp profile + T^2 scaling + m^{-6} scaling + inverse-temperature heating. Semiclassical gravity shares T^2 and $1/T$ heating but differs in mass scaling.

Model	Profile	T scaling	Mass scaling	Heating
IAM	E_q ramp	T^2	m^{-6}	$P \propto 1/T$
Penrose–Diósi	Exponential	None	$m^{-5/3}$	None
CSL	Exponential	None	m^{-1}	$P \propto T$
Graviton emission	Exponential	T^5	Various	$P \propto T^5$
Semiclassical grav.	Exponential	T^2	m^{-2}	$P \propto 1/T$

7 Implications for Quantum Computing

7.1 Current Architectures and the Gravitational Floor

Gravitational decoherence is irrelevant for all current quantum computing platforms. Superconducting qubits ($\sim 10^{-20}$ kg), trapped ions ($\sim 10^{-25}$ kg), and photonic systems ($E_G = 0$) all have τ_{IAM} far exceeding any practical timescale. IAM does not predict that current quantum computers will fail.

However, IAM identifies a fundamental engineering target distinct from isolation. The *isolation paradigm*: decoherence = noise $\rightarrow$ engineer silence $\rightarrow$ move to the Moon. The *IAM paradigm*: decoherence = irreversible information production $\rightarrow$ engineer reversibility $\rightarrow$ design better gates on Earth. As environmental isolation improves, residual decoherence will increasingly be dominated by irreversible thermodynamic interactions in gate operations. At that point, further isolation provides diminishing returns.

7.2 Five Principles for iam-Informed Quantum Computing

Principle 1 — Work with the ramp. Design computations that complete within $t < 0.5 \tau_{\text{IAM}}$, where the decoherence rate is near zero.

Principle 2 — Minimize irreversible interaction. Only irreversible interactions ($Q \geq k_{\text{B}}T \ln 2$) advance E_q . Reversible interactions do not contribute to gravitational decoherence.

Principle 3 — The $\Sigma = 1$ principle. Photons have $E_{\text{G}} = 0$ and $\tau_{\text{IAM}} = \infty$. Photonic qubits are fundamentally immune to gravitational decoherence by the causal structure of spacetime, not engineering.

Principle 4 — The feedback loop as resource. When a qubit decoheres, information is encoded rather than destroyed. Instrumenting the environment to capture this encoding enables computation through controlled decoherence.

Principle 5 — The Landauer floor as constraint. Minimum energy cost per irreversible operation is $k_{\text{B}}T \ln 2$. Optimize to operate as close to this floor as possible.

8 Falsification Criteria

IAM's quantum-scale predictions are falsified if:

1. No gravitational decoherence is detected at 10^{13} – 10^{14} amu with environmental decoherence controlled below the IAM-predicted rate.
2. The mass scaling exponent is inconsistent with $\alpha = 6$ (IAM) or $5/3$ (Penrose–Diósi).
3. Temperature dependence is inconsistent with T^2 .
4. The decoherence profile is exponential, not E_q ramp.
5. Coherence times exceed τ_{IAM} for a given mass.

Falsification of the quantum predictions would not automatically falsify the cosmological predictions ($\mu < 1$, $\Sigma = 1$), which could arise from a different microscopic mechanism with the same macroscopic consequences. The two test regimes are complementary: cosmology tests the cumulative effect; the laboratory tests the underlying cause.

9 Connection to Cosmological Validation

The decoherence rate $E_G^3/(\hbar k_B^2 T^2 \ln 2)$ at laboratory scales is the same physical process that, accumulated over $\sim 10^{11}$ galaxies and 13.8 Gyr, produces the informational entropy that modifies cosmic expansion. The laboratory measures the rate per event; cosmology measures the cumulative effect.

At cosmological scales: $E(a) = \exp(1 - 1/a)$, modifying the matter-sector perturbation equations and producing $\mu < 1$, $\Sigma = 1$. At quantum scales: $E_q(\eta) = \exp(1 - 1/\eta)$, governing the decoherence profile of individual mesoscopic systems. Same equation, same thermodynamics, same physics at every scale.

Simultaneous positive results — Euclid detecting $\mu < 1$ and optomechanical experiments measuring the E_q ramp at the predicted τ_{IAM} — would close the loop from quantum decoherence to cosmic acceleration.

10 Conclusion

We have derived the quantum analog $E_q(\eta) = \exp(1 - 1/\eta)$ of the cosmological activation function $E(a) = \exp(1 - 1/a)$, extending IAM from cosmic to quantum scales. The IAM decoherence timescale $\tau_{IAM} = \hbar k_B^2 T^2 \ln 2 / E_G^3$ contains zero free parameters and produces four unique experimental signatures. Modified Lindblad dynamics under the time-dependent E_q rate show a measurable purity difference peaking at $\eta \approx 0.23$, providing the optimal discrimination measurement time. Detection forecasts show 5σ discrimination above 3.7×10^{-15} kg, with the optimistic detection timeline converging with Euclid's cosmological test in

the early 2030s.

The missing piece was always information.

Acknowledgments

The author thanks the developers of CAMB, MGCAMB, and Cobaya for making cosmological validation accessible to independent researchers. All MCMC chains, configuration files, modified Fortran source code, and analysis scripts are publicly archived at <https://github.com/hmahaffeyges/IAM-Validation>. DOI: [10.17605/OSF.IO/KCZD9](https://doi.org/10.17605/OSF.IO/KCZD9).

References

- [1] H. W. Mahaffey, “Constraints on Late-Time $f\sigma_8$ Suppression from $\mu < 1$, $\Sigma = 1$: Planck 2018 and Large-Scale Structure,” submitted (2026).
- [2] H. W. Mahaffey, IAM Validation Repository (2026). <https://github.com/hmahaffeyges/IAM-Validation>. DOI: [10.17605/OSF.IO/KCZD9](https://doi.org/10.17605/OSF.IO/KCZD9).
- [3] T. Jacobson, “Thermodynamics of Spacetime: The Einstein Equation of State,” *Phys. Rev. Lett.* **75**, 1260 (1995).
- [4] R. Penrose, “On Gravity’s Role in Quantum State Reduction,” *Gen. Relativ. Gravit.* **28**, 581 (1996).
- [5] L. Diósi, “A Universal Master Equation for the Gravitational Violation of Quantum Mechanics,” *Phys. Lett. A* **120**, 377 (1987).
- [6] R. Landauer, “Irreversibility and Heat Generation in the Computing Process,” *IBM J. Res. Dev.* **5**, 183 (1961).
- [7] A. Bérut et al., “Experimental Verification of Landauer’s Principle Linking Information and Thermodynamics,” *Nature* **483**, 187 (2012).

- [8] M. A. Ciampini et al., “Erasure of a Nonequilibrium Memory Beyond Landauer’s Bound Using Levitated Optomechanics,” *Phys. Rev. Research* **7**, 043321 (2025).
- [9] M. Rossi et al., “Quantum Delocalization of a Levitated Nanoparticle,” *Phys. Rev. Lett.* **135**, 083601 (2025).
- [10] U. Delić et al., “Cooling of a Levitated Nanoparticle to the Motional Quantum Ground State,” *Science* **367**, 892 (2020).
- [11] L. Neumeier et al., “Fast Quantum Interference of a Nanoparticle via Optical Potential Control,” *PNAS* **121**, e2306953121 (2024).
- [12] R. Kaltenbaek et al., “MAQRO: Testing Quantum Physics in Space,” *Exp. Astron.* **34**, 123 (2012).
- [13] arXiv:2512.02838, “Experimental Blueprint for Distinguishing Decoherence from Objective Collapse” (2026).
- [14] Planck Collaboration, “Planck 2018 Results. VI. Cosmological Parameters,” *Astron. Astrophys.* **641**, A6 (2020).
- [15] W. H. Zurek, “Decoherence, Einselection, and the Quantum Origins of the Classical,” *Rev. Mod. Phys.* **75**, 715 (2003).